

3D PMS — Project Islands

Design Plan · A project-management system for WorkOS
where the work you do becomes a 3D world you build.

3D PMS — "Project Islands" Design Plan

A project-management system for WorkOS where work you do becomes a 3D world you build. A task is a structure under construction; as the task progresses the structure rises; when the task is done the structure is finished. This doc is the plan — no code yet.

1. The core idea

A task is a construction site.

- Start a task -> a structure begins rising on your project's island.
- Tick off the task's subtasks -> the structure grows, floor by floor.
- Finish the task -> the structure is complete and polished.
- Put it on hold -> construction freezes under a tarp.
- Cancel it -> it collapses to rubble.

You can read a whole project's health from across the room: a skyline of finished buildings, cranes on the work-in-progress, storms over the overdue, tarps on the paused.

2. The building blocks (hierarchy)

Layer	What it is	In the 3D world
Workspace	The WorkOS company	The whole archipelago
Project	One initiative	One island / world
Team	A group of people	A color + a district on the island
Member	A person (with a role)	A builder / hard-hat color
Task	One piece of work	One structure
Subtask	A checklist item	One visible build step

Switch project -> fly to a different island. Each island shows only its own tasks.

3. Data model (the backbone)

Plain shapes; field names are indicative.

Project

```
id, name, description
theme           // visual style of the island (grassland / desert / coast)
team_ids[]      // teams working this project
review_required // does a build need manager sign-off before Done?
status          // active | archived
created_at
```

Team

```
id, name
color           // used to tint this team's structures / district
member_ids[]
```

Member

```
id, name, email
role           // manager | member | viewer
color
```

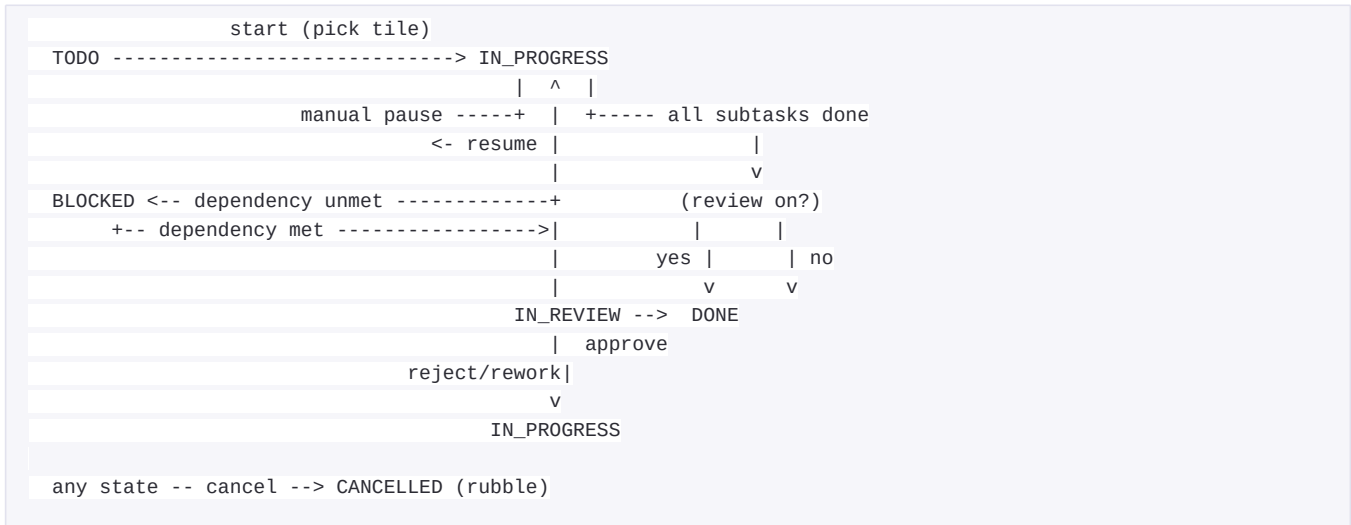
Task (the heart)

```
id, title, description
project_id
team_id
assignee_id, assigner_id
category       // building | nature | vehicle | landmark -> which asset family
size           // 15 | 30 | 45 | 60 (story points) -> how grand the final build
priority       // low | normal | high | urgent
status        // todo | in_progress | on_hold | blocked | in_review | done | cancelled
subtasks[]    // [{ id, title, done }] - progress = done/total
progress       // derived: % of subtasks done
depends_on[]    // task ids that must finish first
world_x, world_z // tile position on the island
sprint_id      // which build-season it belongs to
due_date
reviewer_id
created_at, updated_at
```

Sprint

```
id, project_id, name, goal (points), starts_at, ends_at, status
```

4. Task lifecycle (the state machine)



Key rules: - **Progress = done subtasks ÷ total subtasks**. Ticking a box grows the build. - **Review is optional per project**. If off, the last subtask completes the task straight to Done. If on, 100% moves it to In Review for manager sign-off. - **Reject ≠ demolish**. Rejecting at review sends the task *back to In Progress* ("rework"), scaffolding returns, the building survives. - **Demolish/rubble is only for Cancelled tasks**.

5. Task state -> 3D structure (the master mapping)

Status	Progress	What you see on the tile
To Do	0%	Blueprint plot — flat outline + a signpost with the task name
In Progress	1–33%	Foundation + scaffolding, a crane appears
In Progress	34–66%	Walls up, half-built, materials around
In Progress	67–99%	Full shape, scaffolding still wrapped on
On Hold	frozen	Current stage + tarp draped over, colors dimmed, pause icon, crane stopped
Blocked	frozen	Current stage + barrier/warning look, road drawn to the blocking task
In Review	100%	Looks complete + glowing review pin above it
Done	100%	Fully built, polished, celebration (flag / lights / sparkle)
Cancelled	—	Rubble

6. Subtasks = visible construction milestones

Each ticked subtask does something you can see, not just move a number.

A 5-subtask castle: tick 1 -> foundation, tick 2 -> ground floor, tick 3 -> upper floors, tick 4 -> roof, tick 5 -> flags & polish.

- Every tick spawns a dust-puff and raises the structure a step.
- More subtasks than natural stages -> bunch them. Fewer -> bigger jumps per tick.
- The **final** tick always triggers the pretty reveal.

The checklist and the building become the same thing. "Midway" is never arbitrary — it's literally "3 of 5 subtasks, so 3 of 5 of the way up."

7. What gets built: category × size

Two dials decide the final structure.

Category (the manager sets a task-type) -> the asset family:

Work type	Family	15 pts	30 pts	45 pts	60 pts
Feature / infra	Building	hut	cottage	manor	castle
Research / growth	Nature	sapling	tree	grove	great oak
Ops / logistics	Vehicle	cart	wagon	truck	ship
Design / polish	Landmark	lantern	fountain	statue	monument

Size (story points) -> which rung of that ladder, i.e. how grand.

So "a big infra task" = a castle under construction; "a small design task" = a lantern going up.

8. The construction visual strategy (how we make this buildable)

The honest constraint: we cannot hand-model a "half-built castle," "half-built ship," etc. — that's hundreds of assets. So:

- **In-progress look is universal.** While building, every task shows the same kit — scaffolding poles + crane + a wrapped block whose **height grows with progress %**. Each subtask tick bumps the height and puffs dust.
- **The real model appears only at Done**, revealed with a pop.

- For the **code-drawn shapes** (cottage, tree, fountain — we control their geometry) we can grow them smoothly for real.

One reusable "construction costume" everywhere; the beautiful payoff at the finish. Achievable with the asset kits we already have.

9. Special states, in detail

- **On Hold** (manual): stop advancing, drape a tarp, dim the colors, float a pause icon, stop the crane. Progress frozen and not editable until resumed.
 - **Blocked** (usually automatic from [depends_on](#)): freeze + a barrier look + a road connecting to the blocking structure. Clears automatically when the blocker finishes.
 - **Rework** (reject at review): back to In Progress, scaffolding returns, a small note marker. The build is *not* destroyed.
 - **Cancelled**: collapses to rubble. The only path to rubble.
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10. Persistence & sprints (the growing skyline)

- **Completed builds persist** — a project's island becomes a skyline of everything shipped. History you can see.
 - **Focus without clutter**: past-sprint builds **fade into a background skyline**; the current sprint is bright and front-of-stage.
 - Zoom out to admire the whole city built over quarters; zoom in to work.
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11. Health signals (glanceable, no reading required)

- **Weather = schedule health**. On track -> sunny. Due soon -> clouds. Overdue -> a little storm/rain over that structure.
 - A manager flying over the island instantly sees which builds are "under storm."
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12. Richer PMS features

- **Dependencies = roads**. "B needs A" draws a road between their plots; B stays blocked until A is done, then the road lights up.
- **Epics / milestones = districts**. Related tasks cluster into a neighborhood; finishing them all completes a central monument for that epic.
- **Teams = color + district**. Each team's builds carry their color and can occupy a zone of the island.

- **Leaderboard / velocity.** Finished builds earn points; the board shows who's shipping — real sprint velocity, made fun.
- **Site Foreman** (repurpose the old Quartermaster agent): an activity feed — "Asha started construction on Login," "3 builds finished this sprint."

Upside ideas (nice-to-have, add later)

- **Bugs = fires.** A bug filed against a shipped feature makes that feature's building catch fire / crack until the bug task is fixed. Quality problems show up on the skyline.
- **Overload = crowded worksite.** Too many active tasks on one person = a chaotic, over-cranned lot. Managers spot burnout at a glance.

13. The screens (views)

1. **World view** — the island for one project: build, tick subtasks, watch it rise.
2. **Task detail panel** — title, description, subtasks (the progress driver), assignee, due date, dependencies, review controls.
3. **Board / list view** — a normal kanban/list for people who want plain PM, kept in sync with the 3D world.
4. **Portfolio / archipelago view** — all project islands with health + progress rings; click one to dive in. The execs' "how's everything going" screen.
5. **Dashboards** — velocity, leaderboard, sprint burndown.

14. Roles & permissions

Action	Manager	Member	Viewer
Create project / team	Yes	—	—
Assign task (to anyone)	Yes	self only	—
Set task type / size	Yes	—	—
Start build, tick subtasks	Yes	Yes (own)	—
Submit for review	Yes	Yes (own)	—
Approve / reject / cancel	Yes	—	—
View world & dashboards	Yes	Yes	Yes

15. Phased build roadmap

Phase 1 — Projects & teams (foundation). Multiple projects (islands), a project switcher, teams with members. Mostly data + navigation, minimal new 3D. Low risk.

Phase 2 — Progress-driven construction (the wow). Expand statuses, subtasks drive a progress %, structure grows through stages (blueprint -> scaffolding -> done). Place the structure when the task starts.

Phase 3 — Categories & asset ladders. Task type -> asset family; size -> grandeur. On-hold / blocked / rework markers.

Phase 4 — PMS depth. Dependencies + roads, epics as districts, due dates + priority, weather health, board/list view synced with the world.

Phase 5 — WorkOS integration & polish. Swap the local dummy backend for WorkOS's real auth/teams/data. Foreman activity feed into WorkOS notifications. Portfolio view. Optional: bugs-as-fires, overload.

16. Decisions locked in

- Progress is driven by **subtasks** (each tick = a visible build step).
- Manager sets **task-type + size** at creation; assignee picks the **tile** and builds.
- **Reject ≠ demolish** — rejection is a rework loop; only Cancelled tasks turn to rubble.
- Islands **persist**; past sprints fade into a background skyline.
- **Weather** shows schedule health.
- Bugs-as-fires and overload-as-crowding are **upside**, added later.

17. Open questions to settle before Phase 2

- Exactly when does the plot appear — at **assign** (manager drops the plot) or at **start** (assignee places it)? (Leaning: at start.)
- Is **review** on by default, or opt-in per project?
- How big is an island, and what happens when a project outgrows it (expand vs new island / "expansion pack")?
- Do subtasks carry **weights**, or is every subtask an equal share of progress? (Leaning: equal, for simplicity.)

18. Asset gaps to check

- Scaffolding / crane / barrier / tarp pieces for the universal construction look.
- "Done" hero models across all four families and four sizes (some exist in the current kits; audit what's missing).
- Weather props (cloud / rain / sun) and district / road tiles. ````